Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Data Analytics and Visualization with R

TGS Ref No: TGS-2026064475

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 12.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
11.0	Legacy release	Legacy two-day course and PPT used as the source baseline.	Tertiary Infotech Academy
12.0	20 August 2026	Rebuilt for TGS-2026064475; mapped to the current visual deck, ten self-contained R labs and WA/PP instruments.	Dr Alfred Ang

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TABLE OF CONTENTS

Course Overview **4**

Learning Outcomes **4**

Daily Schedule **4**

Day 1: Prepare, summarise and analyse quantitative research data 4

Day 2: Analyse text, visualise evidence and interpret findings 5

Topic-by-Topic Delivery Guide **5**

Topic 01: Data Preparation and Transformation 5

Topic 02: Data Summary 6

Topic 03: Quantitative Data Analysis 6

Topic 04: Qualitative Data Analysis 7

Topic 05: Data Visualization 7

Resources Required **8**

Assessment **8**

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Course Overview

Data Analytics and Visualization with R is a two-day, 16-hour WSQ programme aligned to Analysis of Research Data (HCE-DAT-4008-1.1). Learners progress from controlled preparation through quantitative and qualitative analysis to reproducible visual communication in R.

Course Code	TGS-2026064475
Training Duration	16 hours across 2 days
Assessment	WA 30 minutes; PP 90 minutes
Trainer	Dr Alfred Ang
Mode	Instructor-led classroom with individual R labs and Excel evidence workbooks

Learning Outcomes

- LO1: Prepare and transform data for analysis using R.
- LO2: Summarise data with descriptive statistics, figures, graphs and tables.
- LO3: Perform quantitative data analysis and interpret the findings.
- LO4: Perform qualitative data analysis and determine significance or effect.
- LO5: Create effective data visualisations in R and interpret research findings.

Daily Schedule

Day 1: Prepare, summarise and analyse quantitative research data

Time	Duration	Topic / Activity	Method	Slides
9:00-9:20	20 min	Welcome, AM digital attendance, introductions, outcomes and evidence conventions	Facilitated briefing	—
9:20-10:20	60 min	Topic 1: research-data workflow, tidy structure, import, missingness, transformation, joins and reconciliation	Concept briefing, editable chart examples and questioning	Slides 17-41
10:20-10:30	10 min	Tea break	Break	—
10:30-12:30	120 min	Labs 1-2: prepare dengue data; join, reshape and aggregate vaccination data	Demonstration, individual R practice and evidence debrief	Slides 37-40
12:30-13:30	60 min	Lunch break	Lunch	—
13:30-14:20	50 min	Topic 2: data types, centre, spread, grouped summaries, distributions and sampling	Concept briefing, editable chart examples and questioning	Slides 42-66
14:20-15:20	60 min	Labs 3-4: descriptive statistics, grouped tables and basic plots	Demonstration, individual R practice and evidence debrief	Slides 62-65
15:20-15:30	10 min	Tea break	Break	—
15:30-16:20	50 min	Topic 3: correlation, regression, hypothesis tests, effect size and ANOVA	Concept briefing, editable chart examples and questioning	Slides 67-91
16:20-17:40	80 min	Labs 5-6: correlation/regression and tests/group comparison	Demonstration, individual R practice and evidence debrief	Slides 87-90

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17:40-18:00	20 min	Day 1 evidence review, Q&A and PM digital attendance	Reflection and Q&A	—

Scheduled learning and assessment time excluding lunch: 480 minutes (8 hours).

Day 2: Analyse text, visualise evidence and interpret findings

Time	Duration	Topic / Activity	Method	Slides
9:00-9:15	15 min	Day 1 recap and AM digital attendance	Facilitated briefing	—
9:15-10:15	60 min	Topic 4: qualitative evidence, text preparation, tokenisation, stop words, frequency and coding	Concept briefing, editable chart examples and questioning	Slides 93-117
10:15-10:25	10 min	Tea break	Break	—
10:25-12:10	105 min	Labs 7-8: term-frequency analysis and reproducible word-cloud critique	Demonstration, individual R practice and evidence debrief	Slides 113-116
12:10-12:30	20 min	Topic 5 introduction: grammar of graphics and question-to-chart fit	Concept briefing, editable chart examples and questioning	Slides 118-142
12:30-13:30	60 min	Lunch break	Lunch	—
13:30-14:25	55 min	Topic 5: scales, geoms, facets, accessibility, visual QA and interpretation	Concept briefing, editable chart examples and questioning	Slides 118-142
14:25-15:25	60 min	Labs 9-10: layered visuals and capstone evidence briefing	Demonstration, individual R practice and evidence debrief	Slides 138-141
15:25-15:35	10 min	Tea break	Break	—
15:35-16:00	**25 min**	**Course recap, assessment briefing and assessment digital attendance**	**Individual summative assessment**	**—**
16:00-16:30	**30 min**	**Written Assessment (WA/SAQ), open book**	**Individual summative assessment**	**—**
16:30-18:00	**90 min**	**Practical Performance (PP), open book**	**Individual summative assessment**	**—**

Scheduled learning and assessment time excluding lunch: 480 minutes (8 hours).

Topic-by-Topic Delivery Guide

Topic 01: Data Preparation and Transformation · Slides 17-41

Research-data workflow · import/export · tidy structure · cleaning · joins · transformation

- **Question before code:** Define the decision, population, variables and unit of observation before transforming data.
- **Tidy structure:** Each variable is a column, each observation is a row and each value occupies one cell.
- **Typed imports:** Dates, factors, text and numbers need deliberate types so analysis behaves predictably.
- **Transformation pipeline:** Select, filter, mutate, join, reshape and aggregate as visible reproducible stages.
- **Missingness is information:** Distinguish structurally absent, unknown and invalid values before removal or imputation.
- **Reconciliation:** Prove row counts, keys and material totals before and after every transformation.

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Lab 1: Prepare and Transform Dengue Surveillance Data • Slides 37-38

Alignment: LO1 · A1. R script: lab-01-data-preparation.R. Excel evidence: lab-01-data-preparation.xlsx. Learner output: A reproducible clean dengue time-series dataset, transformation log and reconciliation evidence. Acceptance: The script runs without error, output rows reconcile to input less documented exclusions, dates are valid, and the clean CSV and plot are created.

Lab 2: Join, Reshape and Aggregate Vaccination Data • Slides 39-40

Alignment: LO1 · A1 · A8. R script: lab-02-join-reshape.R. Excel evidence: lab-02-join-reshape.xlsx. Learner output: A joined long-form vaccination dataset, unmatched-key report, annual summary and transformation audit. Acceptance: The left join preserves all source rows, unmatched keys are zero or documented, and total doses reconcile across input, long and wide summaries.

Topic 02: Data Summary • Slides 42-66

Data types · descriptive statistics · distribution · grouped summaries · basic plots and tables

- **Variable type drives method:** Nominal, ordinal, discrete and continuous data support different summaries and charts.
- **Centre:** Mean and median answer different questions when the distribution is skewed or contains outliers.
- **Spread:** Range, interquartile range, variance and standard deviation quantify variability.
- **Grouped summaries:** Compare meaningful sub-populations while preserving denominators and sample sizes.
- **Sampling variability:** A statistic changes from sample to sample; uncertainty is part of the result.
- **Review accuracy:** Recompute, reconcile and inspect distributions before accepting a summary.

Lab 3: Summarise Centre, Spread and Distribution • Slides 62-63

Alignment: LO2 · A2 · A8. R script: lab-03-descriptive-statistics.R. Excel evidence: lab-03-descriptive-statistics.xlsx. Learner output: A descriptive-statistics table, distribution plot, outlier review and accuracy checklist. Acceptance: R and Excel checks agree, units and sample size are stated, outliers remain traceable, and the chosen centre measure matches the observed distribution.

Lab 4: Build Grouped Tables and Basic Plots • Slides 64-65

Alignment: LO2 · A2 · A8. R script: lab-04-grouped-summary.R. Excel evidence: lab-04-grouped-summary.xlsx. Learner output: Frequency/proportion tables, grouped numerical summaries, bar and box plots, and a denominator audit. Acceptance: Counts reconcile to the dataset, proportions sum to 100% within rounding, every grouped statistic includes n, and the written comparison matches both tables and plots.

Topic 03: Quantitative Data Analysis • Slides 67-91

Correlation · regression · hypothesis testing · effect size · ANOVA · interpretation

- **Correlation:** Measures strength and direction of association; it does not by itself establish causation.
- **Regression:** Models an outcome as a function of one or more predictors under explicit assumptions.
- **Hypothesis testing:** Compares observed evidence with a pre-specified null model and significance threshold.
- **Effect size:** Quantifies how large the difference or relationship is in practical terms.
- **ANOVA:** Tests whether group means differ beyond what within-group variability would suggest.
- **Interpretation boundary:** Separate evidence, assumptions, uncertainty and business or research judgement.

Lab 5: Analyse Correlation and Regression • Slides 87-88

Alignment: LO3 · K1 · A3 · A7. R script: lab-05-correlation-regression.R. Excel evidence: lab-05-correlation-regression.xlsx. Learner output: Correlation evidence, fitted regression model, diagnostic plot, prediction and limitations statement. Acceptance: The model output is reproducible, coefficient and R-squared interpretations use correct units, diagnostics are reviewed, and the conclusion avoids causal claims.

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Lab 6: Test Differences and Compare Groups · Slides 89-90

Alignment: LO3 · K1 · A3 · A5 · A7. R script: lab-06-hypothesis-anova.R. Excel evidence: lab-06-hypothesis-anova.xlsx. Learner output: Test-design record, t-test and ANOVA outputs, effect estimates, assumption checks and conclusion. Acceptance: Test choices match the designs, hypotheses were pre-specified, effect estimates accompany p-values, and conclusions state assumptions and practical limitations.

Topic 04: Qualitative Data Analysis · Slides 93-117

Text preparation · tokenisation · stop words · term frequency · word clouds · thematic interpretation

- **Qualitative evidence:** Open-ended responses carry meaning, context and ambiguity that numeric summaries can miss.
- **Text preparation:** Normalise case and punctuation while preserving the source and recording every exclusion.
- **Tokens:** Break text into reproducible analytical units such as words, phrases or coded themes.
- **Stop words:** Remove high-frequency low-information terms using a documented list and domain judgement.
- **Frequency is not importance:** Common terms reveal salience but require context, concordance and representative excerpts.
- **Credible interpretation:** Use a codebook, peer review and traceable excerpts to manage subjectivity.

Lab 7: Prepare Text and Analyse Term Frequency · Slides 113-114

Alignment: LO4 · K1 · A4 · A5. R script: lab-07-text-analysis.R. Excel evidence: lab-07-text-analysis.xlsx. Learner output: Clean token table, stop-word audit, frequency table, channel comparison and interpretation notes. Acceptance: Every token remains traceable to a source record, exclusions are documented, frequency totals reconcile, and the interpretation uses context rather than frequency alone.

Lab 8: Create and Critique a Word Cloud · Slides 115-116

Alignment: LO4 · K1 · A4 · A5 · A7. R script: lab-08-word-cloud.R. Excel evidence: lab-08-word-cloud.xlsx. Learner output: Word-cloud image, ranked-frequency table, sensitivity comparison, excerpt evidence and limitations statement. Acceptance: The visual is reproducible, ranked counts are available, preprocessing sensitivity is documented, and every claim is supported by counts plus traceable excerpts.

Topic 05: Data Visualization · Slides 118-142

Grammar of graphics · quantitative and qualitative plots · customisation · accessibility · findings

- **Question-to-chart fit:** Choose position, length, colour and shape to support the analytical purpose.
- **Grammar of graphics:** Build a plot from data, aesthetics, geoms, statistical transforms, scales, coordinates and facets.
- **Truthful scales:** Units, baselines, aspect ratios and uncertainty must not exaggerate or hide evidence.
- **Accessible design:** Use readable labels, sufficient contrast and redundant encodings beyond colour alone.
- **Layered emphasis:** Guide attention with hierarchy and annotation while keeping the complete context available.
- **Interpret findings:** State the pattern, evidence, uncertainty, decision relevance and limitation.

Lab 9: Build Layered Quantitative Visuals · Slides 138-139

Alignment: LO5 · K2 · A6 · A9. R script: lab-09-ggplot-visuals.R. Excel evidence: lab-09-ggplot-visuals.xlsx. Learner output: A bar chart, faceted distribution plot and annotated scatter plot with a chart-choice rationale. Acceptance: All three plots answer their stated questions, remain legible without colour alone, use truthful scales and carry interpretations supported by the displayed data.

Lab 10: Capstone: Review and Communicate Findings · Slides 140-141

Alignment: LO5 · K2 · A6 · A7 · A8 · A9. R script: lab-10-capstone.R. Excel evidence: lab-10-capstone.xlsx. Learner output: An auditable analysis bundle with control log, statistical results, three decision visuals and a concise

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evidence-led recommendation. Acceptance: The bundle reruns from source data, all controls pass or are explained, visuals agree with statistical outputs, and the recommendation distinguishes evidence from judgement.

Resources Required

- R 4.2 or later and RStudio Desktop.
- Learner Guide and ten numbered lab folders.
- One runnable R script and one Excel intake/control/evidence workbook per lab.
- Supplied synthetic CSV data and writable outputs folders.
- Projector for concept briefings, editable native PowerPoint chart examples and demonstrations.
- LMS/TMS access for courseware, attendance and assessment submission.

Assessment

- Written Assessment (WA/SAQ) — 2 open-ended questions, 30 minutes, open book.
- Practical Performance (PP) — 5 connected R-analysis tasks, 90 minutes, open book.
- Assessment is conducted on Day 2 from 4:00 PM to 6:00 PM.
- Assessment is individual and open book. Learners submit completed papers, R code and requested evidence through the LMS.
- A minimum of 75% attendance and a Competent assessment outcome are required.
- https://lms-tms.tertiaryinfotech.com/

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